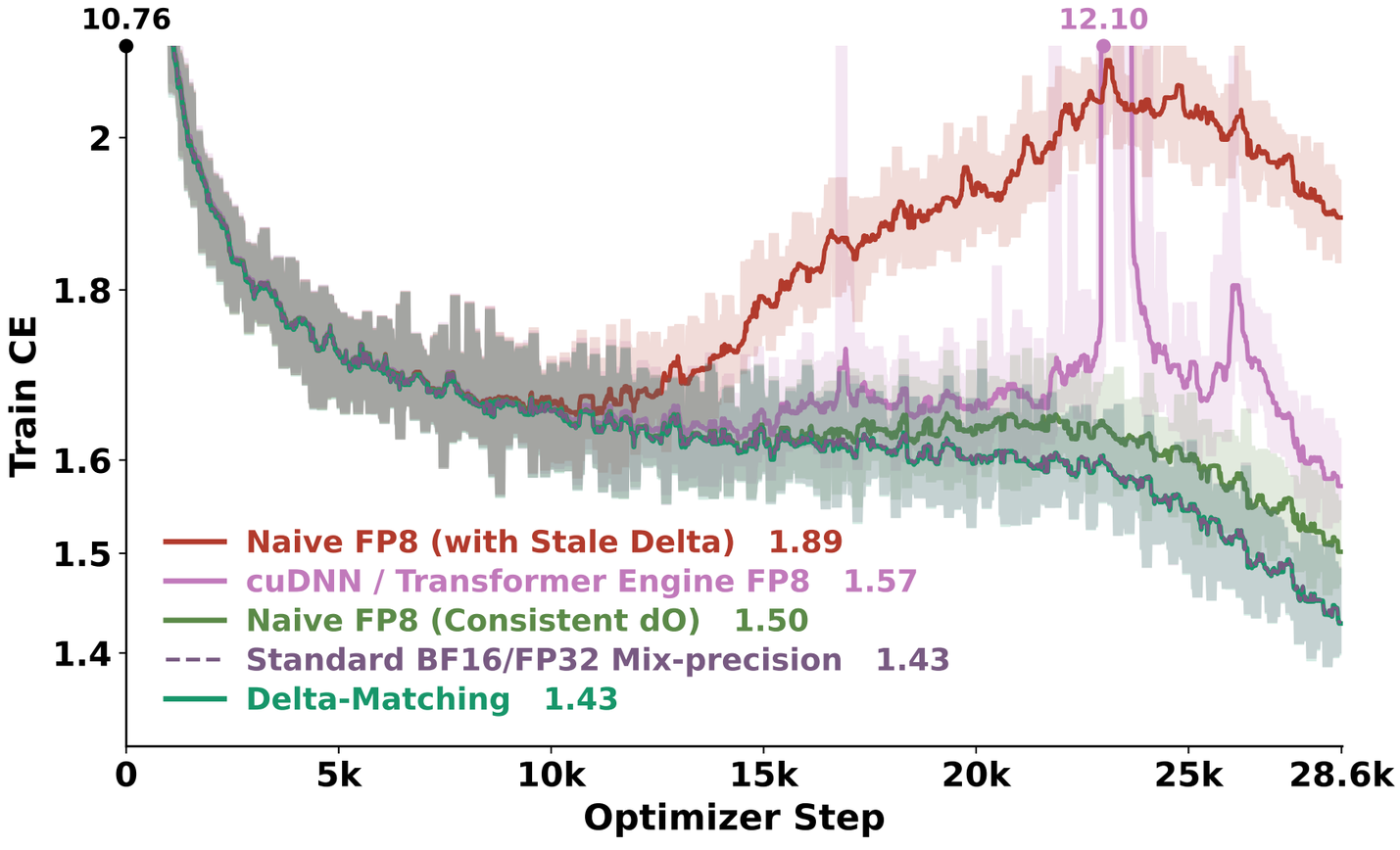
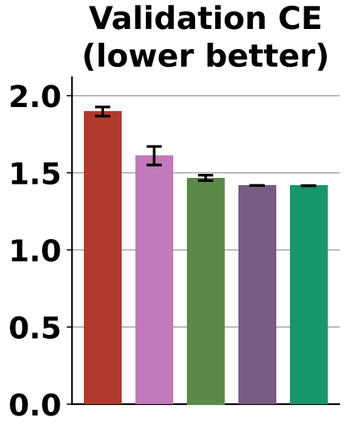
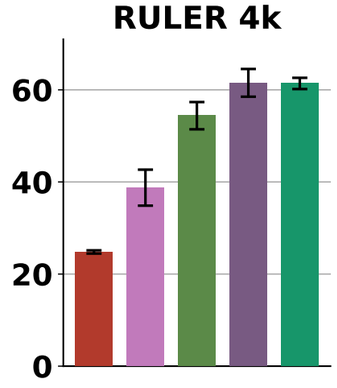
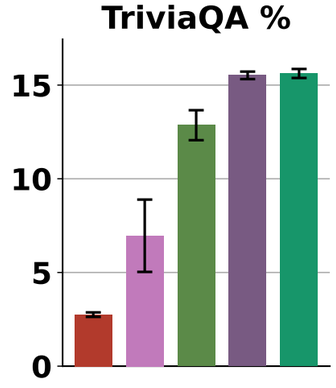
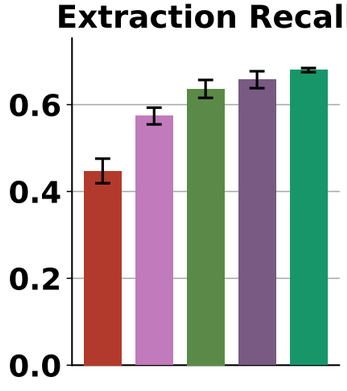
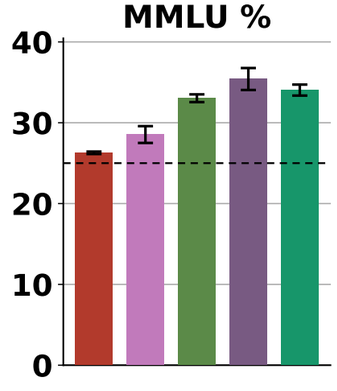

10.76
12.10
Train CE
2
1.8
1.6
1.5
1.4
Naive FP8 (with Stale Delta) 1.89
cuDNN / Transformer Engine FP8 1.57
Naive FP8 (Consistent dO) 1.50
Standard BF16/FP32 Mix-precision 1.43
Delta-Matching 1.43
0
5k
10k
15k
20k
25k
28.6k
Optimizer Step

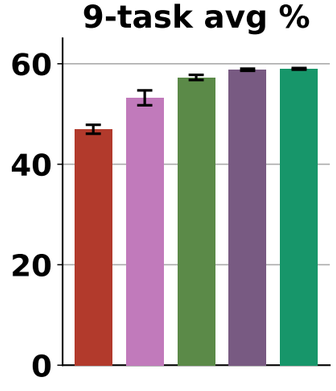

Validation CE
(lower better)
2.0
1.5
1.0
0.5
0.0


RULER 4k
60
40
20
0


TriviaQA %
15
10
5
0


Extraction Recall
0.6
0.4
0.2
0.0


MMLU %
40
30
20
10
0


Commonsense
9-task avg %
60
40
20
0